

Problem-Solution Fit

Date	28 June 2026
Team ID	[Enter Team ID]
Project Name	HDI Predictor
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Problem-Solution Fit Canvas - HDI Predictor

1. CUSTOMER SEGMENT(S) [CS] - Policy makers and development researchers - International development organizations - Economics students and academic researchers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited access to high-compute resources - Missing or sparse country indicator records - Non-standardized data formats from different countries - Lack of machine learning expertise among researchers - Dependence on manual data wrangling	5. AVAILABLE SOLUTIONS PROS & CONS [AS] - Manual Spreadsheet Calculations: Free but slow, error-prone, and static - Traditional UNDP Databases: Authoritative but lagging and lacks forecasting - Machine Learning Models: Fast, scalable, accurate, and adapts to complex trends
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Manual calculations of HDI are slow and error-prone - Calculation errors during socio-economic assessments - Large volume of country-year data to clean and process - Lagging annual UNDP reports prevent timely policy analysis	9. PROBLEM ROOT / CAUSE [RC] - Growing volume of socio-economic parameters - Non-linear mathematical calculations of dimension indices - Subjective thresholds for development tiers - Sparse or incomplete country-year records - Inefficient spreadsheet-based forecasting methods	7. BEHAVIOR + ITS INTENSITY [BE] - Researchers expect instant prediction feedback - Institutions evaluate hundreds of indicator scenarios during policy cycles - Need for high-speed, consistent, and mathematically sound forecasting
3. TRIGGERS TO ACT [TR] - Request for immediate policy impact assessments - Need for rapid scenario forecasting - Researcher demand for instant index predictions - Requirement to automate data wrangling pipelines	10. YOUR SOLUTION [SL] Develop a country Human Development Index (HDI) Prediction System using Machine Learning (Linear Regression, Random Forest and Gradient Boosting) integrated with a Flask web application. The system predicts the HDI score and development tier instantly based on country indicators, improving forecasting accuracy, reducing analysis times, and assisting policy makers in making reliable decisions.	8. CHANNELS OF BEHAVIOR [CH] ONLINE - Research dashboards - Institutional web portals - Open-source code repositories (GitHub) OFFLINE - Academic journals and publications - Direct policy reports - Development planning conferences
4. EMOTIONS BEFORE / AFTER [EM] Before: Frustration with massive spreadsheets, formula errors, and lagging static data After: Instant prediction feedback, high model transparency, confidence in simulation decisions		